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precog-data-intake: Automated discovery, validation, and optimized download management of Earth system model data from Earth System Grid Federation nodes.

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Summary

`precog-data-intake` is a Python command-line-interface (CLI) software for automated discovery, checking, validation, and download of Earth System Model outputs from Earth System Grid Federation (ESGF) archives. The software is designed for research workflows that require reproducible access to large, distributed climate-model datasets and is particularly aimed at bulk screening of Earth system archives before downstream analysis. The software builds on inherited ESGF access functionality from `intake-esgf` (Collier et al., 2026), while introducing interactive CLI workflows for archive interrogation, file-availability checks, grid-consistency cross-checks, temporal validation, export of search-result in tabular formats, and optimised download management of shortlisted Earth System Model data products.

Statement of need

Modern Earth System science workflows frequently rely on climate-model data distributed across ESGF nodes. Although ESGF provides a federated infrastructure for searching and accessing these archives, practical research workflows often require additional automation to determine whether a given combination of variables, experiments, ensemble members, and grid configurations is both scientifically suitable and operationally downloadable before substantial time is spent retrieving files.

This issue is especially important for CMIP6-style analyses in which researchers may need to confirm that the same Earth System Model provides paired pre-industrial, historical and ssp-scenario simulations, that multiple variables of interest are available on compatible grids for downstream analyses, and that temporal coverage is continuous across archived files. Manual inspection of catalogue search results can become slow, repetitive, and error-prone when screening many candidate models or variables across multiple ESGF nodes.

Among its features, `precog-data-intake` provides an interactive workflow for shortlisting Earth System Model datasets that satisfy compound criteria across experiments and variables. The software validates temporal coverage, grid consistency, and file availability across published ESGF archives before download. Users can export search results as tabular summaries, inspect and refine shortlisted datasets, and initiate batch downloads or trigger file integrity checks with locally assigned output paths. Retrieved data are then organized into a directory structure suitable for reproducible downstream analysis. This workflow is particularly useful when researchers need to screen many candidate models and variables before selecting datasets that are both scientifically appropriate and operationally accessible within their computational and storage constraints.

State of the field

precog-data-intake builds on the ESGF catalogue node sweeping implementation from intake-esgf (Collier et al., 2026), but targets a different level of the workflow: rather than focusing only on data access primitives and caching data-responses to in-memory use, it provides an interactive command-line environment for ample discovery, screening, validating and managing downloads of published ESGF archive products that satisfy scientific constraints relevant to Earth system and ocean biogeochemistry applications. The software is intended for situations in which researchers need to move efficiently from broad discovery and catalogue searches to a smaller, analysis-ready subset of model output that satisfies practical and scientific constraints.

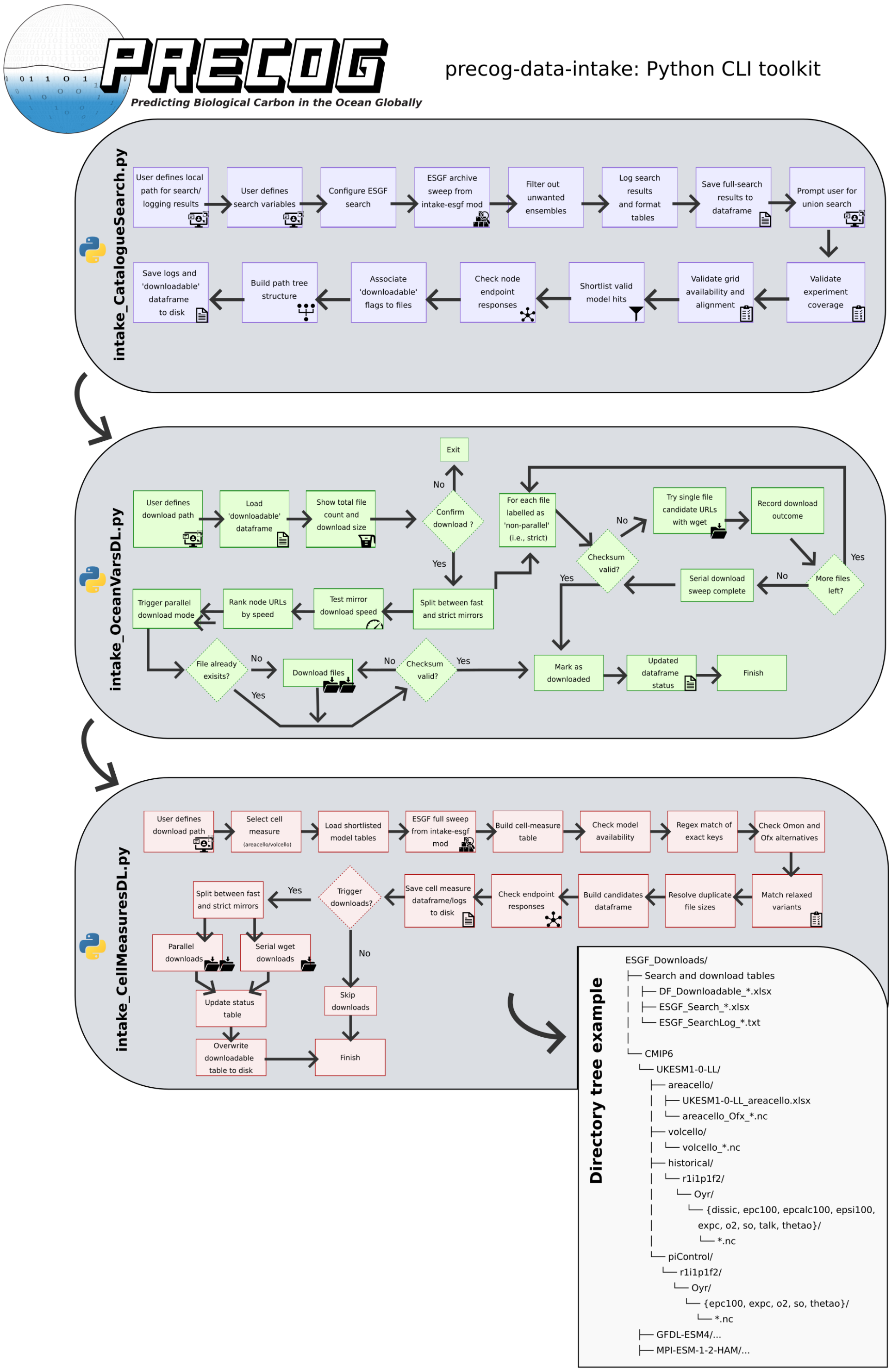
Key functionality

The software provides several features tailored to archive-scale Earth system data workflows:

- automated ESGF node-based searches using project, activity, experiment, frequency, variable, and grid filters (inherited from `intake-esgf`);
- interactive user prompts for download paths and variable selections within CLI workflows;
- bulk screening of CMIP6 archive holdings for paired `piControl` and `historical` availability;
- validation of continuity in date stamps for CMIP6 pre-industrial and historical simulations;
- checks for consistent availability of model output across compatible grids such as `gn` and `gr`;
- conditional multi-variable screening for models that satisfy compound search criteria, including cases where several required variables must be available simultaneously;
- export of tabular ESGF catalogue search results for easy inspection, search snapshot logging, and reproducible shortlist generation;
- parallel URL pre-checks with verification of server responses and flagging of shortlisted outputs as downloadable,
- download manager with parallel transfers that select responsive ESGF node locations and include retry or integrity checks for corrupted files
- in cases where the same file is available across many nodes, the fastest download option is prioritised based on the user's connection
- retrieval of ocean grid-cell measure variables such as `areacello` and `volcello` for selected models, together with saved ESGF archive snapshots for later inspection.

Software design and workflow overview

`precog-data-intake` implements a staged workflow for archive-scale ESGF data discovery and retrieval. Rather than moving directly from catalogue search to cached download, the software separates archive interrogation, shortlist generation, downloadability checks, and file retrieval into distinct command-line steps, allowing users to inspect and validate intermediate results before proceeding. In a typical workflow, the user first specifies a parent download directory and one or more target variables. The catalogue search stage then queries ESGF holdings for matching CMIP6 products, filters results to retain scientifically relevant combinations such as paired `piControl` and `historical` simulations, and exports tabular search summaries for inspection. Subsequent stages verify whether shortlisted files are reachable on remote nodes, assign local destination paths, and download both target variables and required supporting grid-cell measures such as `areacello` and `volcello`. This staged design is intended to improve transparency and reproducibility in archive-based Earth system workflows. By treating search results, validation outputs, and downloadable file lists as explicit intermediate artifacts, the software supports both interactive use and later auditing of dataset selection decisions.



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Research impact and applications

A representative use case is the identification of CMIP6 models that simultaneously provide both pre-industrial and historical outputs for ocean biogeochemical variables such as expc and `epc100`, together with auxiliary or supporting variables and the associated grid-cell measures required for downstream analyses. The repository accompanying this software includes a workflow example. demonstrating this type of archive screening and retrieval process.

This is particularly relevant for ocean biogeochemistry and carbon-cycle studies for example, where analyses often depend on coherent combinations of physical and biogeochemical fields rather than isolated variables (e.g., retrieval of carbonate system fields as well as ocean state physical variables). In such cases, the time spent screening archive holdings, checking consistency, and organizing downloads can be substantial, and purpose-built automation improves both efficiency and reproducibility.

Therefore, `precog-data-intake` fills a workflow gap between catalogue access and scientific analysis. The software capabilities support workflows in which data access is itself a significant part of the scientific process, particularly when analyses depend on assembling coherent, scientifically consistent, and analysis-ready ensemble subsets of Earth system model output from distributed archives.

AI usage disclosure

AI assistance was used solely to improve readability of selected software documentation (Claude Sonnet 5 by Anthropic). No AI tools were used to design, implement, or validate the software. All AI-assisted documentation was reviewed and edited by the authors, who retain full responsibility for its accuracy and content. Authors made all the core design and architectural decisions.

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Collier, N., Grover, M., Stachelek, J., Huard, D., & Andela, B. (2026). *Intake-esgf* (Version v2026.6.4). Zenodo. https://doi.org/10.5281/zenodo.20544597